

chapter 7 Reporting and Interpreting Cost of Goods Sold and Inventory

Financial Accounting

12e

Libby • Libby • Hodge

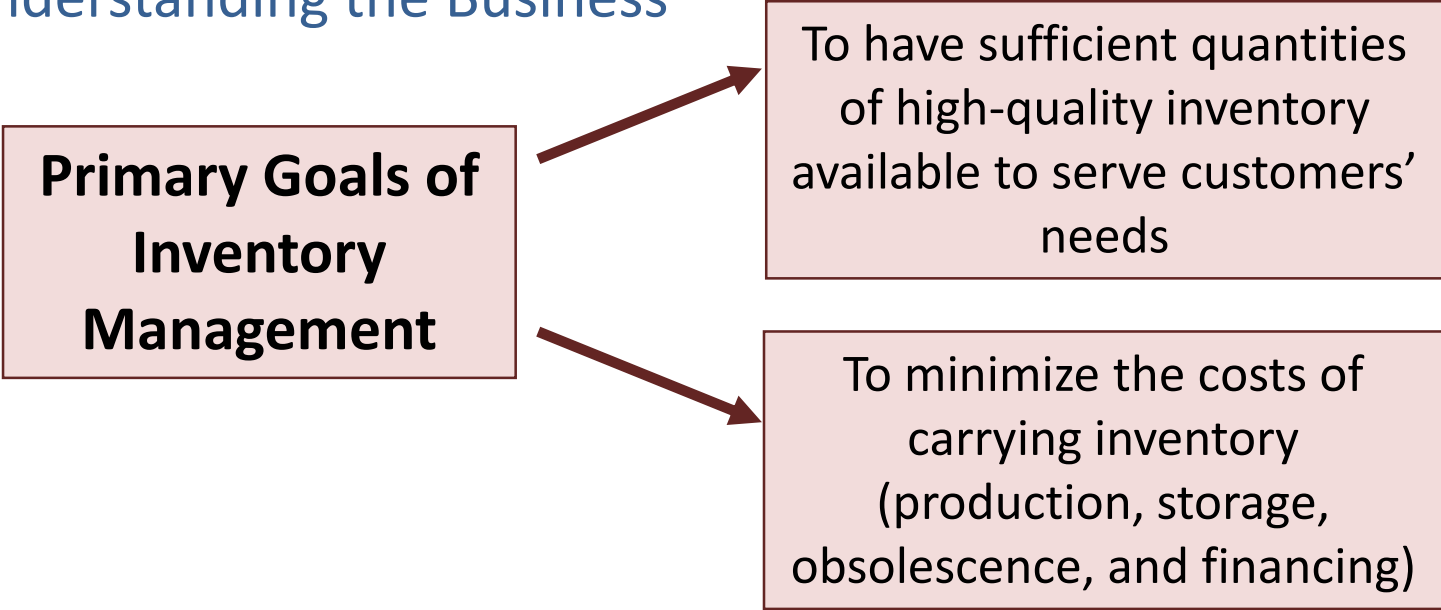
Learning Objectives

After studying this chapter, you should be able to:

- 7-1** Apply the cost principle to identify the amounts that should be included in inventory and cost of goods sold for typical retailers, wholesalers, and manufacturers.
- 7-2** Report inventory and cost of goods sold using the four inventory costing methods.
- 7-3** Decide when the use of different inventory costing methods is beneficial to a company.
- 7-4** Report inventory at the lower of cost or net realizable value.
- 7-5** Understand methods for controlling inventory and analyze the effects of inventory errors on financial statements.
- 7-6** Evaluate inventory management using the inventory turnover ratio and analyze the effects of inventory on cash flows.

Understanding the Business

Primary Goals of Inventory Management



To have sufficient quantities of high-quality inventory available to serve customers' needs

To minimize the costs of carrying inventory (production, storage, obsolescence, and financing)

The accounting system plays three roles in the inventory management process.

- 1) provides accurate information for the financial statements and tax returns.
- 2) provides up-to-date information on inventory quantities and costs to facilitate ordering and manufacturing decisions.
- 3) provides the information needed to help protect these important assets from theft and other misuse.

Exhibit 7.1 Income Statement and Balance Sheet Excerpts

HARLEY-DAVIDSON, INC. Condensed Statements of Operations for the HDMC Segment (Partial) (in thousands)* 12 Months Ended			
	Dec. 31, 2023	Dec. 31, 2022	Dec. 31, 2021
Net Sales	\$4,844,594	\$ 4,887,672	\$4,504,434
Cost of Goods Sold	<u>3,278,052</u>	<u>3,359,799</u>	<u>3,204,907</u>
Gross Profit	\$ 1,566,542	\$ 1,527,873	\$ 1,299,527

HARLEY-DAVIDSON, INC. Consolidated Balance Sheet (Partial) (in thousands)*		
	Dec. 31, 2023	Dec. 31, 2022
Assets		
Current assets:		
Cash and cash equivalents	\$1,533,806	\$ 1,433,175
Accounts receivable, net	267,200	252,225
Finance receivables, net	2,113,729	1,782,631
Inventories, net	<u>929,951</u>	<u>950,960</u>
Restricted cash	104,642	135,424
Other current assets	<u>214,401</u>	<u>196,238</u>
Total current assets	\$ 5,163,729	\$4,750,653

*Harley-Davidson's statements have been simplified for purposes of our discussion.

Learning Objective 7-1

7-1 Apply the cost principle to identify the amounts that should be included in inventory and cost of goods sold for typical retailers, wholesalers, and manufacturers.

Items Included in Inventory

Merchandisers

Merchandise
Inventory

Manufacturers

Raw
Materials
Inventory

Work in
Process
Inventory

Finished
Goods
Inventory

Inventory, a current asset, is tangible property:
(1) held for sale in the normal course of business or
(2) used to produce goods or services for sale.

Inventory Disclosures

Harley-Davidson's recent inventory note reports the following:

HARLEY-DAVIDSON, INC.

Notes to Consolidated Financial Statements

2. ADDITIONAL BALANCE SHEET AND CASH FLOW INFORMATION

Inventories, net (in thousands)

	2023	2022
Raw materials and work in process	\$ 389,221	\$ 331,380
Motorcycle finished goods	514,964	549,041
Parts and accessories and apparel	150,844	187,039
Inventory at lower of FIFO cost or net realizable value	1,055,029	1,067,460

Costs Included in Inventory Purchases

Inventory is initially recorded at cost.

Inventory cost includes the costs to bring an article to usable or salable condition and location.

- + Invoice price
- + Freight-In (*freight charges to deliver items to company warehouse*)
- + Inspection costs
- + Preparation costs
- Purchase returns and allowances
- Purchase discounts
- = **Total inventory cost**

Company should cease accumulating purchase costs when the raw materials are **ready for use** or when the merchandise inventory is **ready for shipment**.

Costs related to selling the inventory should be included in selling, general, and administrative expenses.

Applying the Materiality Constraint in Practice



FINANCIAL ANALYSIS

Incidental costs such as inspection and preparation costs are **not material** and do not have to be assigned to the inventory cost.

Thus, for practice reasons, most companies use the invoice price, less returns and discounts, to assign costs to materials or merchandise and record other incidental features as a separate cost that is reported as an expense.

Most companies use the following method to determine inventory cost:

Invoice Price	XX
Less: Returns	- XX
Less: Discounts	- XX
<u>Total Inventory Cost</u>	<u>XX</u>

Exhibit 7.2

Flow of Inventory Costs

**STAGE 1: PURCHASING/
PRODUCTION ACTIVITIES**

A. MERCHANDISER



B. MANUFACTURER

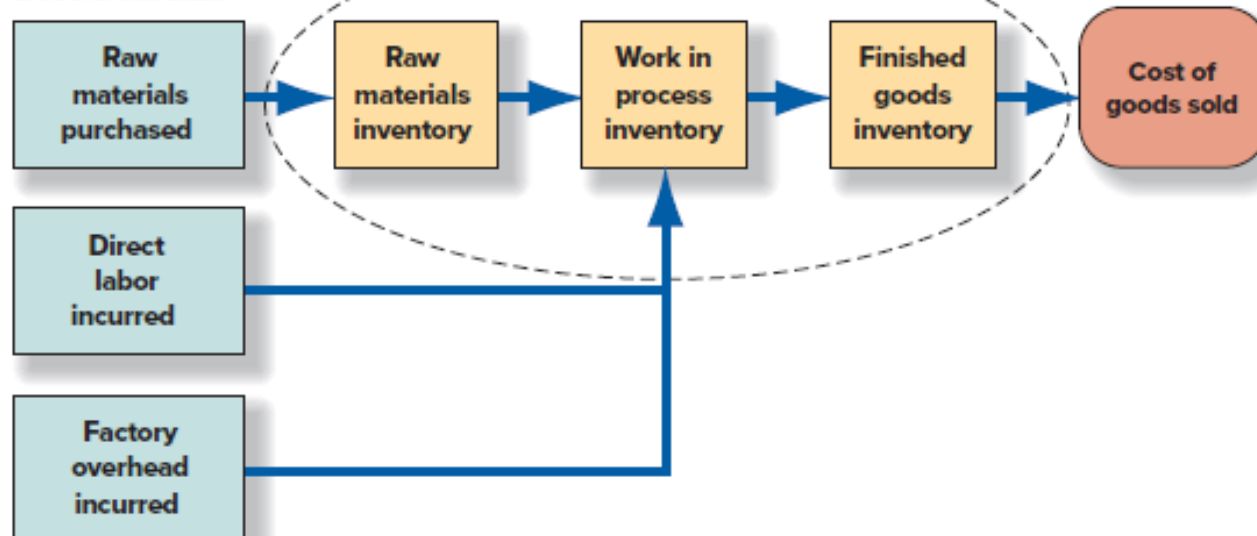


Exhibit 7.3 (1 of 3)

Cost of Goods Sold for Merchandise Inventory

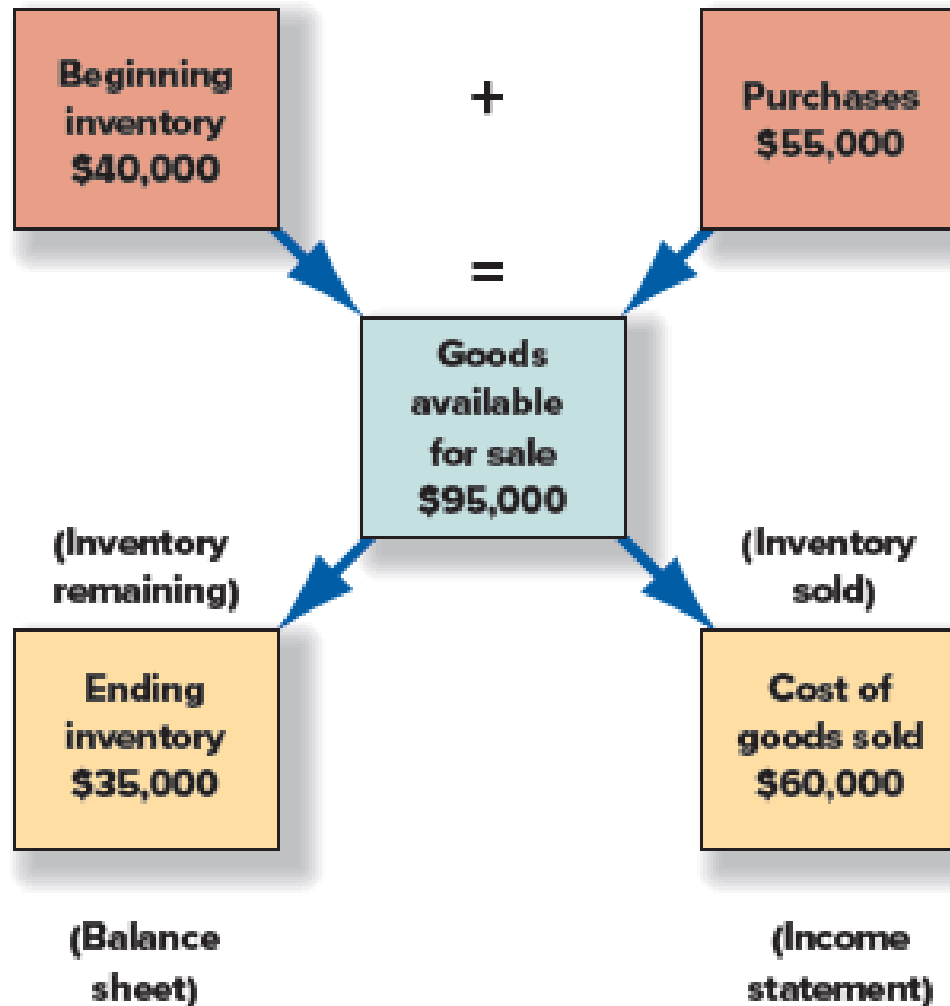


Exhibit 7.3 (2 of 3)

Calculating Cost of Goods Sold

	Beginning inventory
+	Purchases of merchandise during the year
	Goods available for sale
-	Ending inventory
	Cost of goods sold

Cost of goods sold equation: $BI + P - EI = CGS$



Be sure to include **beginning inventory** in all cost of goods sold calculations.

Exhibit 7.3 (3 of 3)

Calculating Cost of Goods Sold

Assume that Harley-Davidson began the period with \$40,000 worth of Motorclothes in beginning inventory, purchased additional merchandise during the period for \$55,000, and had \$35,000 left in inventory at the end of the period. These amounts are combined as follows to compute cost of goods sold of \$60,000:

Beginning inventory	\$40,000
+ Purchases of merchandise during the year	55,000
Goods available for sale	95,000
– Ending inventory	35,000
Cost of goods sold	<u>\$60,000</u>

Merchandise Inventory (A)			
Beginning inventory	40,000		
Add: Purchases of inventory	55,000	Deduct: Cost of goods sold	60,000
Ending inventory	<u>35,000</u>		

Perpetual and Periodic Inventory Systems

The cost of cost of goods sold and ending inventory can be determined by using one of two different inventory systems: perpetual or periodic.

Perpetual

Purchase transactions are recorded directly in an inventory account.

Sales require two entries in the record: (1) the sale and (2) the cost of goods sold.

Used by most companies!
(We'll discuss this throughout the rest of this textbook)

Periodic

A up-to-date record of inventory is maintained during the year.

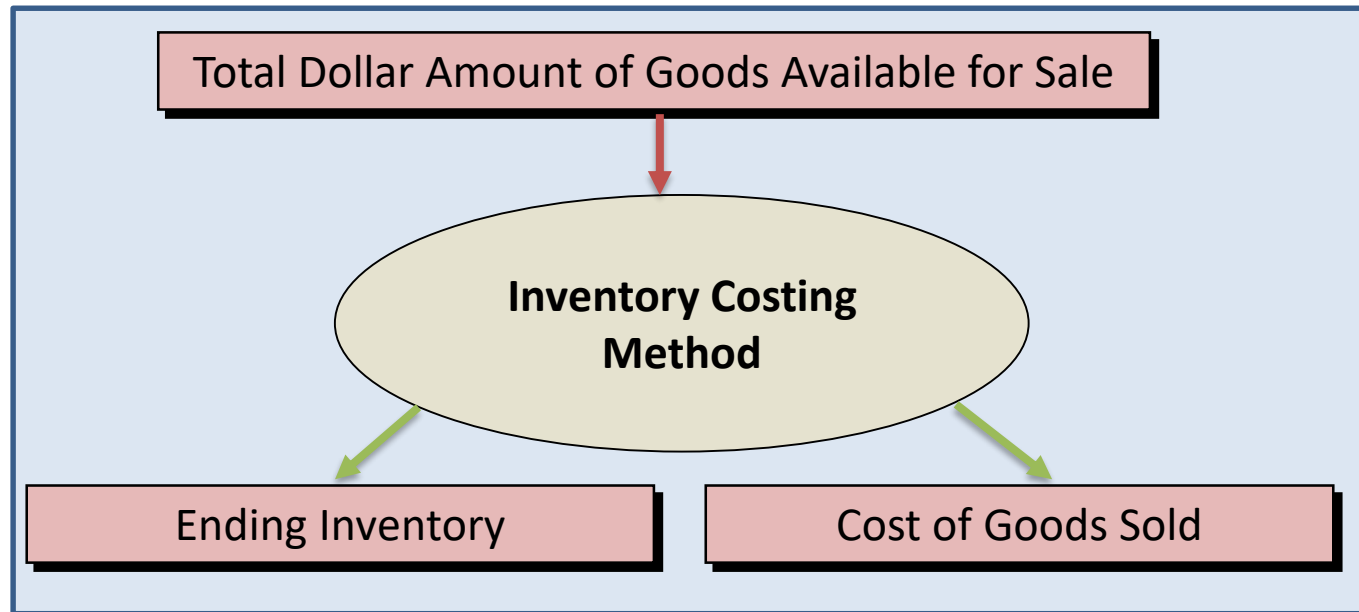
Require one entry to record the sale. Cost of goods sold is calculated at the end of the period.

Disadvantage:
Inventory information is only available at the end of the period.

Learning Objective 7-2

7-2 Report inventory and cost of goods sold using the four inventory costing methods.

Inventory Costing Methods



Inventory Costing Methods

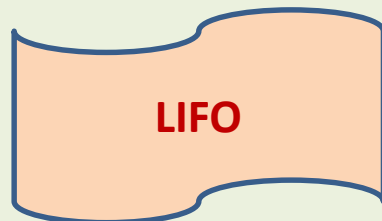
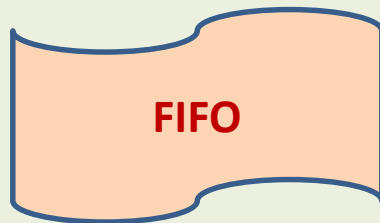
1. Specific identification
2. First-in, first-out (FIFO)
3. Last-in, first-out (LIFO)
4. Average cost

The four inventory costing methods are **alternative ways to assign the total dollar amount of goods available for sale** between ending inventory and cost of goods sold.

Specific Identification Method

The specific identification method **individually identifies and records the cost of each item sold** as cost of goods sold. This method requires keeping track of the purchase cost of each item. This method is **impractical when large quantities of similar items** are stocked. It can be used for expensive unique items such as houses or fine jewelry.

Therefore, most inventory items are accounted for using one of three cost flow assumptions:



Regardless of the actual flow of goods to customers, firms may use any of the three cost flow assumptions described in this chapter. We discuss the rationale for managers' choices later in the chapter.

Cost Flow Assumptions



Key Point!

The choice of an inventory costing method is **not based on the physical flow** of goods on and off the shelves.

That is why they are called **cost flow assumptions**.

Inventory Example

Assume that a Harley-Davidson dealer had the indicated inventory on hand and transactions during January as follows:

- Jan. 1 Had beginning inventory of two units of a Model A leather jacket at \$70 each.
- Jan. 12 Purchased four units of the Model A leather jacket at \$80 each.
- Jan. 14 Purchased one unit of the Model A leather jacket at \$100.
- Jan. 15 Sold four units of the Model A leather jacket for \$120 each.

Note: Three units remain in ending inventory at the end of the period.

First-In, First-Out Method

The **first-in, first-out method**, frequently called **FIFO**, assumes that the earliest goods purchased (the first ones in) are the first goods sold, and the last goods purchased are left in ending inventory.

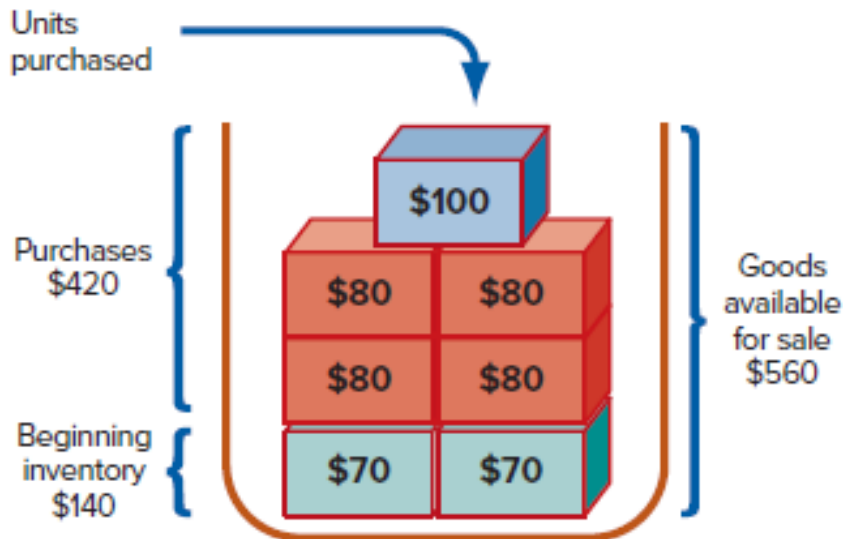
FIFO allocates the **oldest** unit costs **to cost of goods sold** and the **newest** unit costs **to ending inventory**.

Exhibit 7.4

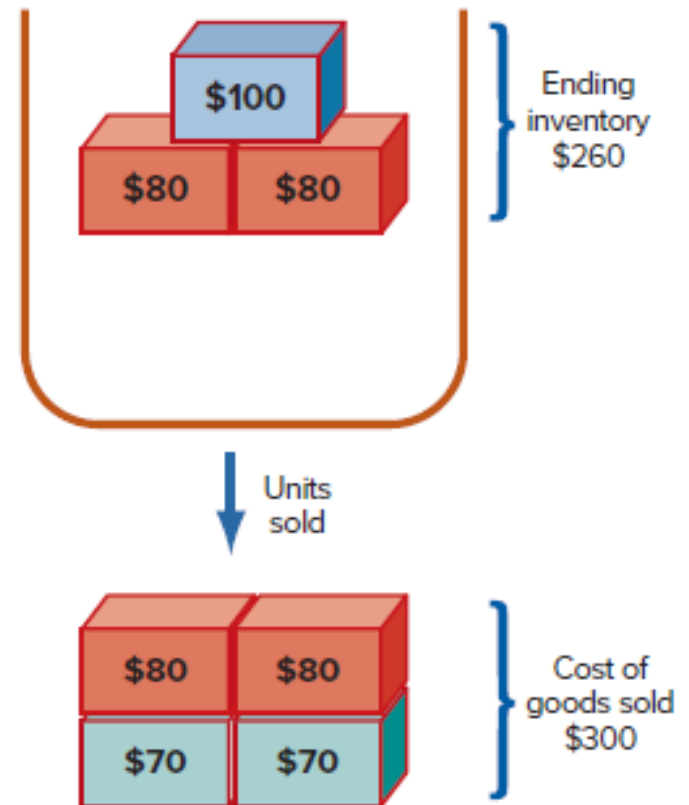
FIFO Inventory Flows

A. FIFO

Step 1: Purchase merchandise

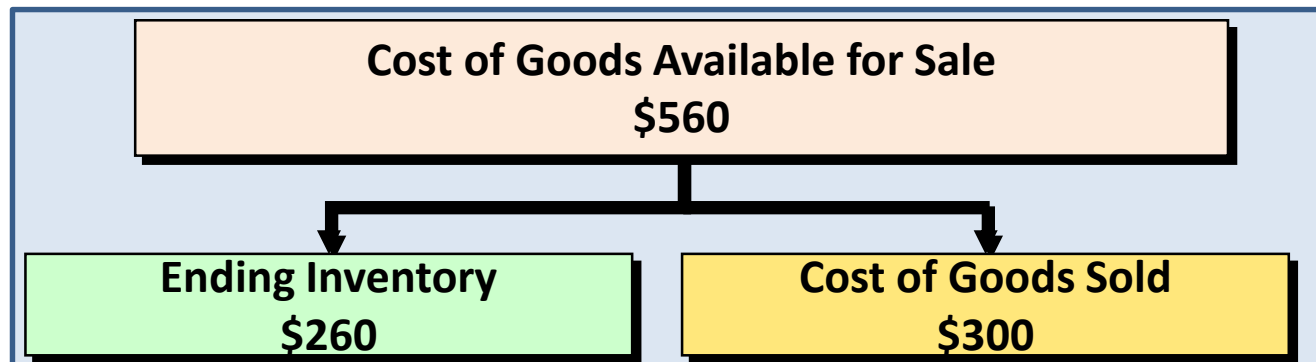


Step 2: Sell merchandise



First-In, First-Out Method

Cost of Goods Sold Calculation (FIFO)		
Beginning inventory	(2 units at \$70 each)	\$140
+ Purchases	(4 units at \$80 each)	320
	(1 unit at \$100)	100
Goods available for sale		560
– Ending inventory	(2 units at \$80 each and 1 unit at \$100)	260
Cost of goods sold	(2 units at \$70 each and 2 units at \$80 each)	<u>\$300</u>



Information:

- ✓ At the beginning of the period, the company had two units in beginning inventory.
- ✓ During the period, the company purchased five units
- ✓ During the period four units were sold.
- ✓ Three units remain in ending inventory at the end of the period.

Last-In, First-Out Method

The **last-in, first-out method**, often called **LIFO**, assumes that the most recently purchased goods (the last ones in) are sold first and the oldest units are left in ending inventory.

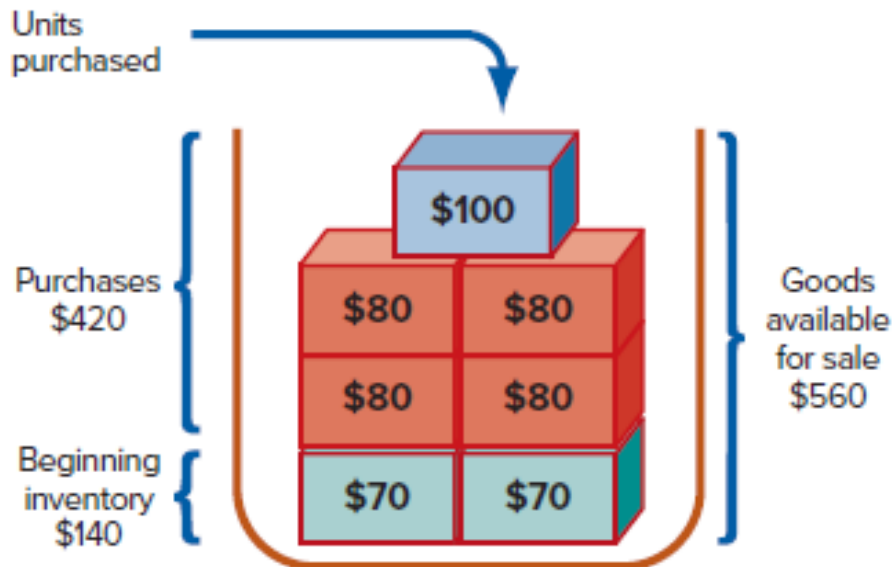
LIFO allocates the **newest** unit costs to **cost of goods sold** and the **oldest** unit costs to **ending inventory**.

Exhibit 7.4

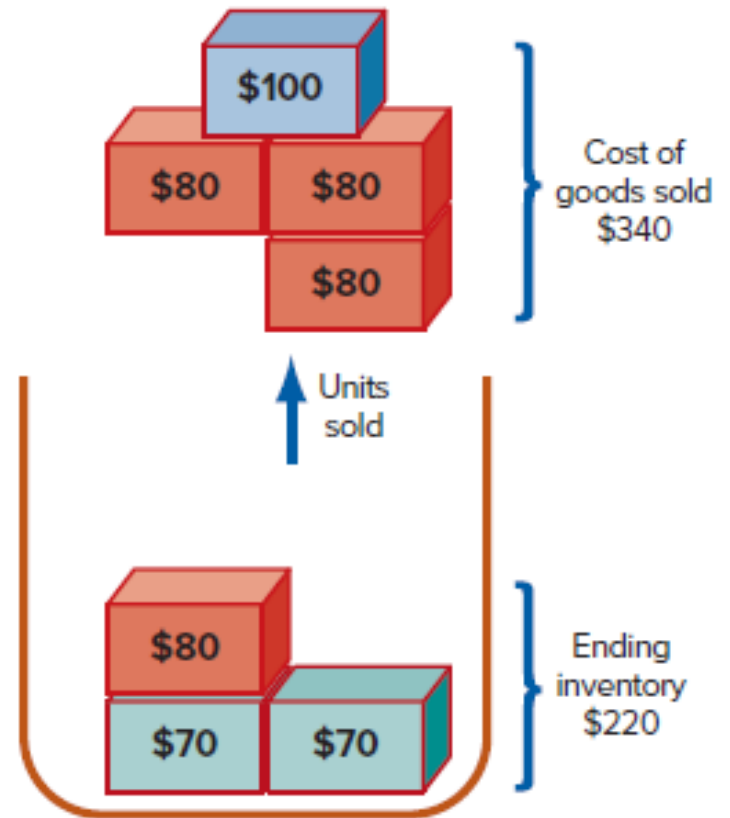
LIFO Inventory Flows

B. LIFO

Step 1: Purchase merchandise

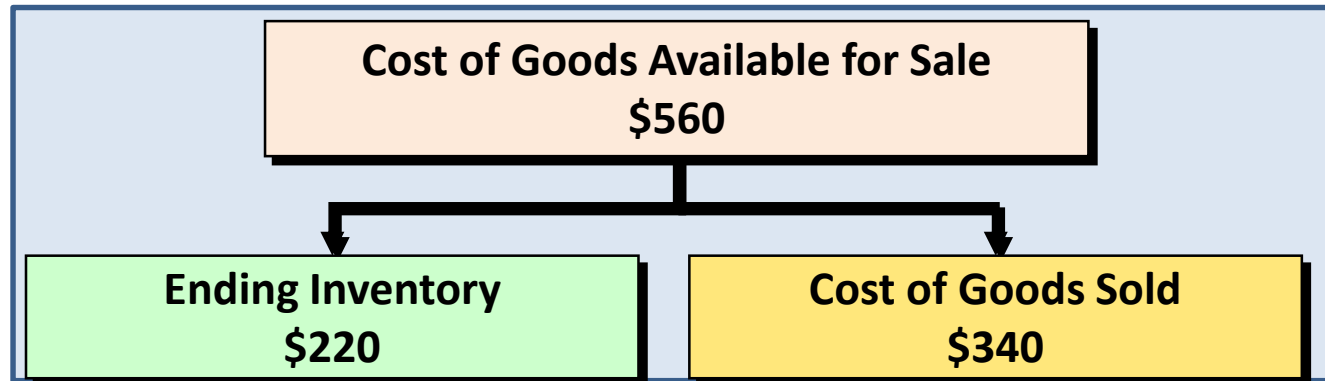


Step 2: Sell merchandise



Last-In, First-Out Method

Cost of Goods Sold Calculation (LIFO)		
Beginning inventory	(2 units at \$70 each)	\$140
+ Purchases	(4 units at \$80 each)	320
	(1 unit at \$100)	<u>100</u>
Goods available for sale		560
– Ending inventory	(2 units at \$70 each and 1 unit at \$80)	<u>220</u>
Cost of goods sold	(3 units at \$80 each and 1 unit at \$100)	<u><u>\$340</u></u>



Information:

- ✓ At the beginning of the period, the company had two units in beginning inventory.
- ✓ During the period, the company purchased five units
- ✓ During the period four units were sold.
- ✓ Three units remain in ending inventory at the end of the period.

Last-In, First-Out Method

The LIFO cost flow assumption is the exact opposite of the FIFO cost flow assumption:

	FIFO	LIFO
Cost of goods sold on income statement	Oldest unit costs	Newest unit costs
Inventory on balance sheet	Newest unit costs	Oldest unit costs

Average Cost Method (1 of 2)

The **average cost method** (weighted average cost method) uses the weighted average unit cost of the goods available for sale for both cost of goods sold and ending inventory.

The weighted average unit cost of the goods available for sale is computed as follows:

Number of Units	×	Unit Cost	=	Total Cost
2	×	\$ 70	=	\$ 140
4	×	\$ 80	=	320
1	×	\$100	=	100
<u>7</u>				<u>\$560</u>

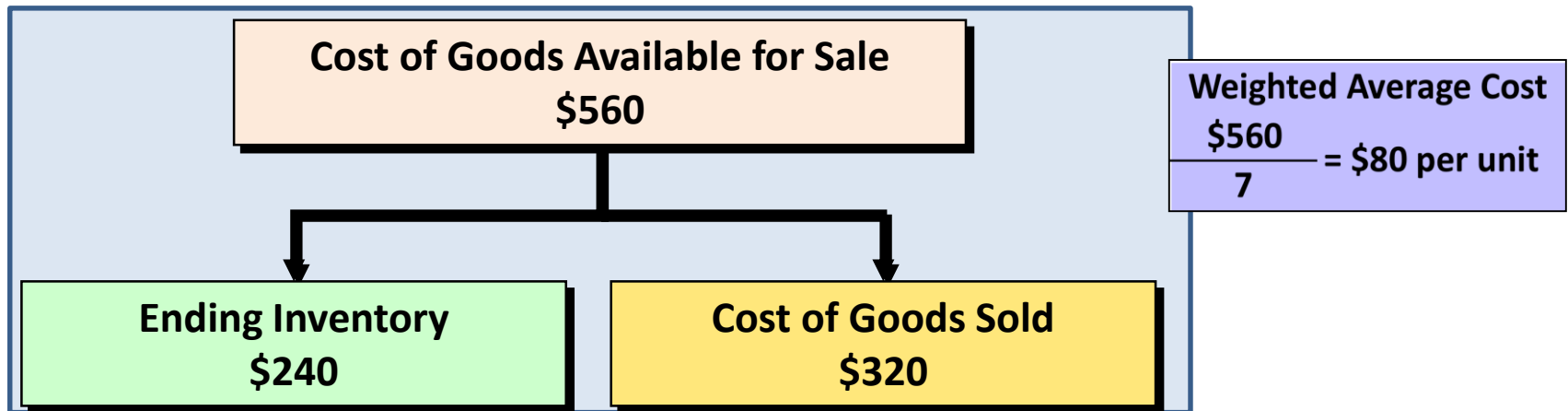
$$\text{Average cost} = \frac{\text{Cost of Goods Available for Sale}}{\text{Number of Units Available for Sale}}$$

$$\text{Average cost} = \frac{\$560}{7 \text{ Units}} = \$80 \text{ per Unit}$$

Cost of goods sold and ending inventory are assigned the **same weighted average cost per unit** of \$80.

Average Cost Method (2 of 2)

Cost of Goods Sold Calculation (Average Cost)		
Beginning inventory	(2 units at \$70 each)	\$140
+ Purchases	(4 units at \$80 each)	320
	(1 unit at \$100)	100
Goods available for sale	(7 units at \$80 average cost each)	560
– Ending inventory	(3 units at \$80 average cost each)	240
Cost of goods sold	(4 units at \$80 average cost each)	<u>\$320</u>



Information:

- ✓ At the beginning of the period, the company had two units in beginning inventory.
- ✓ During the period, the company purchased five units
- ✓ During the period four units were sold.
- ✓ Three units remain in ending inventory at the end of the period.

Perpetual Inventory Systems and Cost Flow Assumptions in Practice

Most companies make numerous purchases and sales of the same inventory item throughout the accounting period.

How do companies keep track of inventory costs using a perpetual inventory system?

- 1) Virtually all accounting systems that keep track of the costs of individual items do so on a FIFO or average cost basis, regardless of the cost flow assumption used for financial reporting.**
- 2) Companies that wish to report under LIFO convert the outputs of their perpetual inventory system to LIFO with an adjusting entry at the end of each period.** By waiting until the end of the period to calculate this LIFO adjustment, LIFO ending inventory and cost of goods sold are calculated as if all purchases during the period were recorded before cost of goods sold was calculated and recorded.

LIFO and International Comparisons



INTERNATIONAL PERSPECTIVE

While U.S. GAAP allows companies to choose among FIFO, LIFO, and average cost inventory methods, International Financial Reporting Standards (IFRS) currently prohibit the use of LIFO.

GAAP also allows different inventory accounting methods to be used for different types of inventory items.

IFRS requires that the same method be used for all inventory items that have a similar nature and use.

These differences can create comparability problems when one attempts to compare companies across international borders.

Each individual country's tax laws determine the acceptability of different inventory methods for tax purposes.

Exhibit 7.5 (1 of 2)

Financial Statement Effects of Inventory Costing Methods

	FIFO	LIFO	Average Cost
Effect on the Income Statement			
Sales	\$480	\$480	\$ 480
Cost of goods sold	<u>300</u>	<u>340</u>	<u>320</u>
Gross profit	180	140	160
Other expenses	<u>80</u>	<u>80</u>	<u>80</u>
Income before income taxes	100	60	80
Income tax expense (25%)	<u>25</u>	<u>15</u>	<u>20</u>
Net income	<u>\$ 75</u>	<u>\$ 45</u>	<u>\$ 60</u>
Effect on the Balance Sheet			
Inventory	<u>\$260</u>	<u>\$220</u>	<u>\$ 240</u>

Exhibit 7.5 (2 of 2)

Financial Statement Effects of Inventory Costing Methods

Increasing Costs: Normal Financial Statement Effects	FIFO	LIFO
Cost of goods sold on income statement	Lower	Higher
Net income	Higher	Lower
Income taxes	Higher	Lower
Inventory on balance sheet	Higher	Lower
Decreasing Costs: Normal Financial Statement Effects	FIFO	LIFO
Cost of goods sold on income statement	Higher	Lower
Net income	Lower	Higher
Income taxes	Lower	Higher
Inventory on balance sheet	Lower	Higher

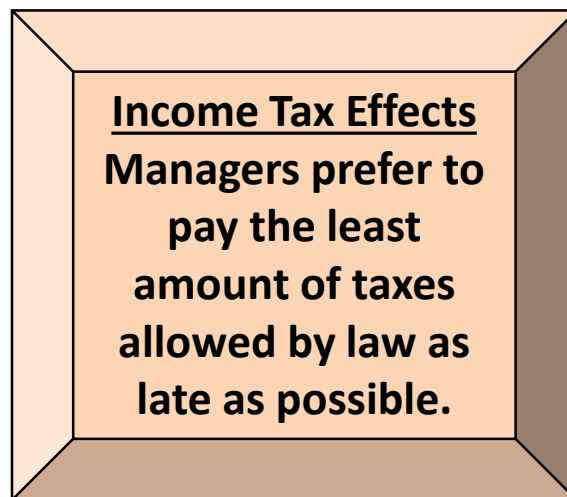
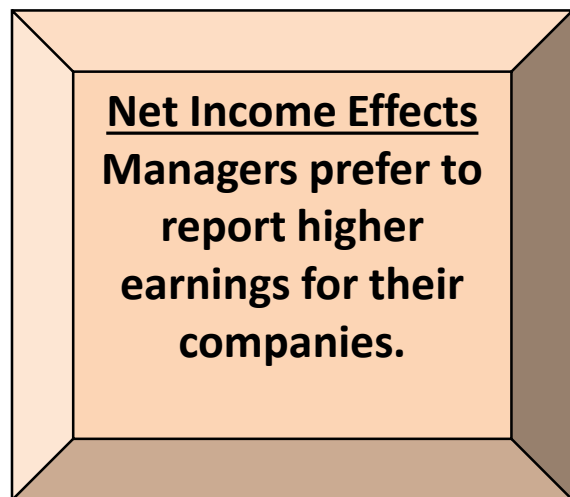
- ❖ **When unit costs are rising**, LIFO produces lower net income and a lower inventory valuation than FIFO.
- ❖ **When unit costs are declining**, LIFO produces higher net income and higher inventory valuation than FIFO.

Learning Objective 7-3

7-3 Decide when the use of different inventory costing methods is beneficial to a company.

Managers' Choice of Inventory Methods (1 of 2)

Most managers choose accounting methods based on two factors:



Any conflict between the two motives is normally resolved by choosing one accounting method for external financial statements and a different method for preparing tax returns. **HOWEVER: If LIFO is used on the U.S. income tax return, it also must be used to calculate inventory and cost of goods sold for the financial statements.**
This is called the **LIFO Conformity Rule**.

Managers' Choice of Inventory Methods (2 of 2)

Increasing Cost Inventories

LIFO is used on the tax return because it normally results in lower income taxes

For inventory located in countries that do not allow LIFO for tax purposes, or do not have a LIFO conformity rule, companies most often use FIFO or average cost to report higher income on the income statement

Decreasing Cost Inventories

FIFO is most often used for **both** the tax return and financial statements. FIFO produces the lowest tax payments for companies with decreasing cost inventories.



In the United States, managers normally choose LIFO to account for their increasing cost inventories to reduce income taxes. They choose FIFO or average cost for their decreasing cost and non-U.S. inventories.

REMINDER: Regardless of the physical flow of goods, a company can use any of the inventory costing methods.

Accounting rules require companies to apply their accounting methods on a consistent basis over time. A company is not permitted to use LIFO one period, FIFO the next, and then go back to LIFO. A change in method is allowed only if the change will improve the measurement of financial results and financial position.

LIFO and Conflicts between Managers' and Owners' Interests



A QUESTION OF ETHICS

The selection of an inventory method can have significant effects on the financial statements.

Company managers may have an incentive to select a method that is not consistent with the owners' interests. For example, during a period of rising prices, using LIFO may be in the interest of the owners (reduce tax liability), however managers may prefer LIFO (typically higher profits) if their compensation is based on profits.

A well-designed compensation plan should align managers' interests with owners' interests.

A manager who selects an accounting method that is not optimal for the company solely to increase his or her compensation is engaging in **questionable ethical behavior.**

Learning Objective 7-4

7-4 Report inventory at the lower of cost or net realizable value.

Valuation at Lower of Cost or Net Realizable Value

Inventories should be measured initially at their purchase cost.

When the **net realizable value** of goods in ending inventory falls below cost, these goods must be assigned a unit cost equal to their net realizable value.

This rule is known as measuring inventories at the **lower of cost or net realizable value** (lower of cost or market).

Net realizable value (NRV) = sales price less costs to sell

Lower of Cost or Net Realizable Value is based on the **conservatism** constraint, which requires companies to avoid overstating assets and income.

This is particularly important for two types of companies:

- 1) High-technology companies**
- 2) Companies that sell seasonal goods**

Net Realizable Value

- If the net realizable value of the inventory is lower than original cost, the company would make a “**write-down**” **entry** to reduce the inventory balance to net realizable value.
 - Companies recognize a “**holding**” **loss** in the period in which the net realizable value of an item drops below original cost.
 - The loss is the difference between the purchase cost and the lower net realizable value.
 - The loss is recorded in the period in which the net realizable value drops, rather than in the period the item is sold.
 - The holding loss is added to cost of goods sold for the period.
- The write-down to lower of cost or NRV **decreases pretax income in the current period** and **increases pretax income in the period of sale** by the same amount.
- *Recognition of **holding gains** on inventory is not permitted by GAAP. No write-down is necessary if the net realizable value is higher than the original cost.*

Valuation at Lower of Cost or Net Realizable Value

Assume that HP had the following items in the current period ending inventory:

Item	Quantity	Cost per Item	Net Realizable Value (NRV) per Item	Lower of Cost or NRV per Item	Total Lower of Cost or Net Realizable Value
Notebooks	1,000	\$250	\$200	\$200	$1,000 \times \$200 = \$200,000$
Printers	400	100	110	100	$400 \times \$100 = 40,000$

The notebooks should be recorded in the ending inventory at the current net realizable value (\$200) because it is **lower** than the cost (\$250). HP makes the following journal entry to record the write-down.

	Debit	Credit
Cost of goods sold (+E, -SE) ($1,000 \times \$50$)	50,000	
Inventory (-A)		50,000

Assets		=	Liabilities	+	Stockholders' Equity	
Inventory	-50,000				Cost of Goods Sold (+E)	-50,000

Because the net realizable value of the printers (\$110) is higher than the original cost (\$100), no write-down is necessary. The printers remain on the books at their cost of \$100 per unit (\$40,000 in total).

Effects of Lower of Cost or NRV Write-Down

The write-down of the notebooks to market produces the following effects on the income statement and balance sheet:

Effects of Lower of Cost or NRV Write-Down	Current Period	Next Period (if sold)
Cost of goods sold	Increase \$50,000	Decrease \$50,000
Pretax income	Decrease \$50,000	Increase \$50,000
Ending inventory on balance sheet	Decrease \$50,000	Unaffected



The write-down to lower of Cost or NRV **decreases pretax income in the current period** and **increases pretax income in the period of sale** by the same amount.

Learning Objective 7-5

7-5 Understand methods for controlling inventory and analyze the effects of inventory errors on financial statements.

Internal Control of Inventory



Any inventory is the asset second most vulnerable to theft. Internal control features focus on safeguarding inventory and providing up-to-date information for management decisions. Examples include:

Separation of responsibilities for inventory accounting and physical handling of inventory.

Maintaining perpetual inventory records.

Storage of inventory in a manner that minimizes risk from theft and damage.

Reconciling perpetual inventory records to periodic physical counts of inventory.

Limiting access to inventory to authorized employees.

Errors in Measuring Ending Inventory

- The measurement of ending inventory quantities and costs affects both the balance sheet (assets) and the income statement (cost of goods sold, gross profit, and net income).
- Therefore, **inventory errors affect both the current year and the next year's income before taxes.**

Assume that ending inventory was overstated by \$10,000 due to a clerical error that was not discovered. This would affect the current year and next year's cost of goods sold as follows:

Current Year		Next Year	
Beginning inventory		Beginning inventory	Overstated \$10,000
+ Purchases of merchandise during the year		+ Purchases of merchandise during the year	
– Ending inventory	Overstated \$10,000	– Ending inventory	
Cost of goods sold	Understated \$10,000	Cost of goods sold	Overstated \$10,000

Learning Objective 7-6

7-6 Evaluate inventory management using the inventory turnover ratio and analyze the effects of inventory on cash flows.

Inventory Turnover (1 of 2)



KEY RATIO ANALYSIS

The primary goals of inventory management are to have enough inventory available to serve customers' while minimizing the costs of carrying inventory. The **inventory turnover ratio** measures the efficiency of inventory management activities.

$$\text{Inventory Turnover} = \frac{\text{Cost of Goods Sold}}{\text{Average Inventory}}$$

Average Inventory is calculated:
(Beginning Inventory + Ending Inventory)

The ratio reflects **how many times average inventory was produced and sold during the period**. A higher ratio indicates that inventory moves more quickly through the production process to the customer, thus reducing storage and obsolescence costs.

Inventory Turnover (2 of 2)



KEY RATIO ANALYSIS

$$\text{Inventory Turnover} = \frac{\text{Cost of Goods Sold}}{\text{Average Inventory}}$$

The 2023 ratio for **Harley-Davidson** (see Exhibit 7-1 for the inputs to the equation):

$$\frac{\$3,322}{(\$929,951 + \$1,000,000)/2} = 3.5$$

COMPARISONS OVER TIME		
Harley-Davidson		
2021	2022	2023
5.2		3.5

COMPARISONS WITH COMPETITORS	
Polaris	Honda Motor
2023	2023
3.8	6.6

Average Days to Sell Inventory



KEY RATIO ANALYSIS

$$\text{Average Days to Sell Inventory} = \frac{365}{\text{Inventory Turnover}}$$

This ratio reflects the **average number of days** it takes a company to produce and deliver inventory to its customers.

Inventory Methods and Financial Statement Analysis

- Converting the Income Statement to FIFO

The choice of a cost flow assumption affects how goods available for sale are allocated to ending inventory and cost of goods sold.

It does not affect the recording of purchases.

Ending inventory will be different under the alternative methods, and, because last year's ending inventory is this year's beginning inventory, beginning inventory also will be different:

Beginning inventory	Different
+ Purchases of merchandise during the year	Same
— Ending inventory	Different
Cost of goods sold	Different

LIFO Reserve or “Excess of FIFO over LIFO”

If we know the differences between a company's inventory valued at LIFO and FIFO for both beginning and ending inventory, we can compute the difference in cost of goods sold.

LIFO reserve (or “Excess of FIFO over LIFO”) refers to the differences between LIFO and FIFO values for beginning and ending inventory.

LIFO Reserve is a **contra-asset** for the excess of FIFO over LIFO Inventory.

The LIFO Reserve is disclosed by LIFO users in their inventory footnotes.

You can adjust the inventory amounts on the balance sheet to FIFO by substituting the FIFO values in the note for the LIFO values.

Alternatively, you can add the LIFO reserve to the LIFO value on the balance sheet to arrive at the same numbers.

Exhibit 7.6

Financial Statement Effects of Inventory Costing Methods

Using Harley-Davidson's LIFO reserve values reported in the footnote presented in Exhibit 7.6, we see that cost of goods sold would have been \$8,578 **lower** had it used FIFO.

Beginning LIFO Reserve (Excess of FIFO over LIFO)	\$ 116,500
– Less: Ending LIFO Reserve (Excess of FIFO over LIFO)	(125,078)
<u>Difference in Cost of Goods Sold under FIFO</u>	<u>\$ (8,578)</u>

HARLEY-DAVIDSON, INC.
Notes to Consolidated Financial Statements
2. ADDITIONAL BALANCE SHEET AND CASH FLOWS INFORMATION
Inventories, net (in thousands)

	2023	2022
...		
Inventory at lower of FIFO cost or net realizable value	1,055,029	1,067,460
LIFO Reserve → Excess of FIFO over LIFO cost	<u>(125,078)</u>	<u>(116,500)</u>
Inventory reported → Total inventories, net on the balance sheet	<u>\$ 929,951</u>	<u>\$ 950,960</u>

Inventory Methods and Financial Statement Analysis

Because FIFO cost of goods sold expense is **lower**, income before income taxes would have been \$8,578 **higher**. Income taxes would be that amount times its 2023 tax rate of 21 percent **higher** had it used FIFO.

Difference in pretax income under FIFO	\$ 8,578
Tax rate	× 0.21
Difference in taxes under FIFO	\$ 1,801

Combining the two effects, net income would be increased by the change in cost of goods sold of \$8,578 and decreased by the change in income tax expense of \$1,801, resulting in an overall increase in net income of \$6,777.

Decrease in Cost of Goods Sold Expense (Income increases)	\$ 8,578
Increase in Income Tax Expense (Income decreases)	(1,801)
Increase in Net Income under FIFO	\$ 6,777

Converting Inventory on the Balance Sheet to FIFO

You can adjust the inventory amounts on the balance sheet to FIFO by substituting the FIFO values in the note (\$1,055,029 and \$1,067,460 for 2023 and 2022, respectively) for the LIFO values (see Exhibit 7.6).

Alternatively, you can add the LIFO reserve to the LIFO value on the balance sheet to arrive at the same numbers.

Inventory and Cash Flows (1 of 3)



FOCUS ON CASH FLOWS

When companies increase production to meet increases in demand, this increases the amount of inventory reported on the balance sheet.

However, when companies overproduce to meet demand for a product:

- They produce too much of the slow-selling item.
- This increases storage costs as well as the interest costs on short-term borrowings that finance the inventory.
- It even can lead to losses if the excess inventory cannot be sold at normal prices.

The cash flow statement often provides the first sign of such problems.

Inventory and Cash Flows (2 of 3)

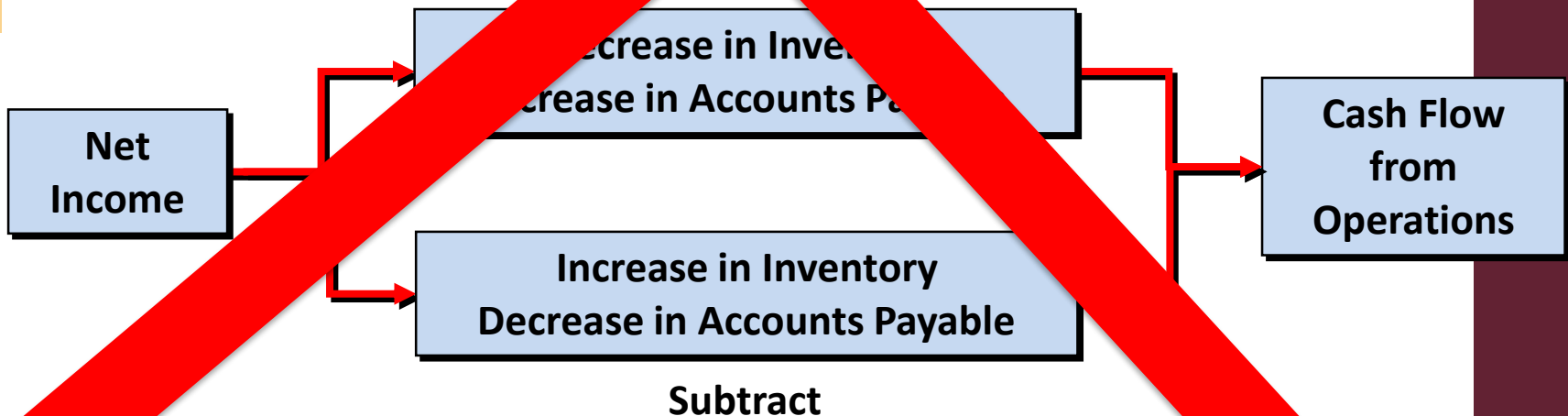


FOCUS ON CASH FLOWS

A change in inventory can have a major effect on a company's cash flow from operations. Because most inventory is purchased on credit (accounts payable), reconciling **cost of goods sold with cash paid to suppliers** requires consideration of the **changes in the Inventory and Accounts Payable accounts.**

Effect on the Statement of Cash

Add



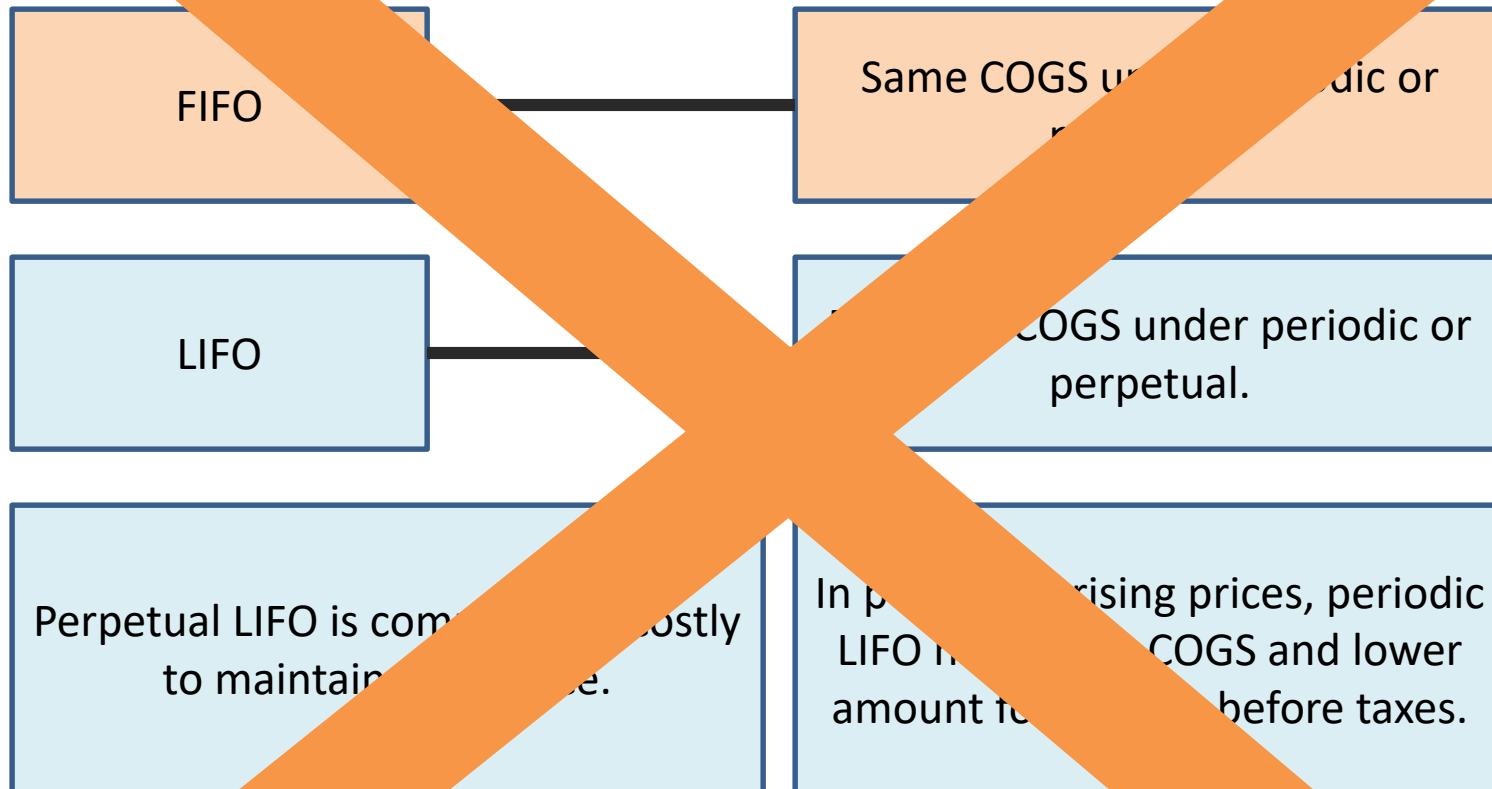
Inventory and Cash Flows (3 of 3)



FOCUS ON CASH FLOWS

Effect on Cash Flows	
Operating activities (indirect method)	
Net income	\$xxx
Adjusted for	
Add inventory decrease	+
or	
Subtract inventory increase	-
Add accounts payable increase	+
or	
Subtract accounts payable decrease	-

Supplement A: FIFO COGS under Periodic and Perpetual Inventory Systems and End of Period Adjustments to



Companies that use perpetual inventory records on a FIFO basis and wish to make an end of period adjusting entry to convert inventory on the balance sheet to a LIFO basis must adjust the cost of goods sold on the income statement to a LIFO basis.

Supplement B: Additional Issues in Measuring Purchases

Purchased goods may be returned if they are damaged or unsatisfactory. **Purchase returns and allowances** reduce the cost of inventory purchases and reduce the amount owed to the vendor. If the invoice was previously paid the return would result in a cash refund.

A **purchase discount** is a cash discount granted for prompt payment of an account (within the discount period). Purchase discounts reduce the cost of inventory.

If payment is made after the discount period, then the full amount of the invoice is due.